

What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence where computers learn from data and make decisions without being explicitly programmed.

Instead of writing every rule manually, we train the machine using data.

Traditional Programming vs Machine Learning

Traditional Programming

Input + Rules $\rightarrow$ Output

Example:

- Marks > 40 $\rightarrow$ Pass
- Marks < 40 $\rightarrow$ Fail

Rules are written manually.

Machine Learning

Input + Output Data $\rightarrow$ Machine Learns Rules

Example:

- Student marks dataset
 - Machine automatically learns patterns
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Simple Definition

Machine Learning allows computers to learn patterns from data and improve automatically through experience.

Everyday Example

YouTube Recommendations

When you watch videos:

- YouTube learns your interests
- Suggests similar videos

This is Machine Learning.

More Examples

Application	ML Usage
Netflix	Movie recommendations
Amazon	Product suggestions
Google Maps	Traffic prediction
Gmail	Spam detection
Instagram	Face filters
Banking	Fraud detection

Why Machine Learning is Important?

- Automation
 - Faster decisions
 - Pattern recognition
 - Predict future outcomes
 - Handle large data
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Difference Between AI, ML, and Deep Learning

AI	ML	Deep Learning
Smart systems	Learning from data	Neural network-based learning

Example:

- AI = Smart Robot
- ML = Robot learns from data
- DL = Robot uses neural networks

Part 2: Real-World Applications of ML

1. Healthcare

- Disease prediction
- Medical image analysis
- Drug discovery

Example:

- Detect cancer from X-rays
-

2. Banking & Finance

- Fraud detection
- Credit score prediction
- Loan approval systems

Example:

- Detect suspicious transactions
-

3. E-commerce

- Product recommendation
- Customer behavior analysis
- Personalized ads

Example:

- Amazon suggesting products
-

4. Transportation

- Self-driving cars
- Route optimization
- Traffic prediction

Example:

- Google Maps traffic alerts
-

5. Education

- Smart learning systems
 - Automatic grading
 - Student performance prediction
-

6. Agriculture

- Crop prediction
 - Weather analysis
 - Smart irrigation
-

7. Social Media

- Face recognition
 - Content recommendation
 - Fake news detection
-

Activity for Students

Ask students:

- “Where do you use ML daily?”

Expected answers:

- Instagram
 - YouTube
 - Google
 - Shopping apps
 - OTT platforms
-

Part 3: Machine Learning Workflow Overview

Complete ML Workflow

1. Collect Data
 2. Clean Data
 3. Prepare Data
 4. Select Model
 5. Train Model
 6. Test Model
 7. Deploy Model
 8. Monitor Model
-

Step 1: Data Collection

Data is gathered from:

- Websites
- Sensors
- Databases
- APIs

Example:

- House price dataset
-

Step 2: Data Cleaning

Remove:

- Missing values
- Duplicate data
- Incorrect values

Example:

- Empty age field
-

Step 3: Data Preparation

Convert data into useful format.

Tasks:

- Feature selection
- Encoding
- Scaling

Step 4: Model Selection

Choose algorithm:

- Linear Regression
- Decision Tree
- KNN

Step 5: Model Training

Machine learns patterns from training data.

Example:

- Predict house price from size

Step 6: Model Testing

Evaluate performance using test data.

Metrics:

- Accuracy
- Precision
- Recall

Step 7: Deployment

Deploy model into:

- Website
- Mobile app
- Software

Step 8: Monitoring

Check:

- Accuracy
- Errors
- Performance

Real Example Workflow

Spam Email Detection

Step	Example
Collect Data	Emails
Clean Data	Remove junk
Train Model	Spam classifier
Test	Check accuracy
Deploy	Gmail spam filter

Simple Dataset Example

```
import pandas as pd

data = {
    "Hours": [1,2,3,4,5],
    "Marks": [20,40,50,70,90]
}

df = pd.DataFrame(data)

print(df)
```

Types of Machine Learning

Main Types

1. Supervised Learning
 2. Unsupervised Learning
 3. Reinforcement Learning
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1. Supervised Learning

Machine learns using:

- Input data
- Correct output labels

Real-Life Example

Student studies with answers.

Applications

- House price prediction
 - Email spam detection
 - Weather prediction
-

Types of Supervised Learning

A. Regression

Predict continuous values.

Example:

- Salary prediction
 - Temperature prediction
-

B. Classification

Predict categories.

Example:

- Spam or Not Spam

- Pass or Fail
-

Supervised Learning Flow

Input Data → Train Model → Predict Output

Example Code: Supervised Learning

Linear Regression Example

```
from sklearn.linear_model import LinearRegression
import numpy as np
```

```
X = np.array([[1],[2],[3],[4],[5]])
y = np.array([20,40,50,70,90])
```

```
model = LinearRegression()
```

```
model.fit(X,y)
```

```
prediction = model.predict([[6]])
```

```
print("Predicted Marks:", prediction)
```

Output:

```
Predicted Marks: [106.]
```

2. Unsupervised Learning

Machine learns without labeled data.

No teacher is available.

Real-Life Example

Grouping students by behavior without knowing categories.

Applications

- Customer segmentation
 - Pattern detection
 - Recommendation systems
-

Clustering

Most common unsupervised method.

Example:

- Divide customers into groups
-

Example

Shopping App:

Group users:

- High spenders
 - Medium spenders
 - Low spenders
-

Unsupervised Learning Flow

Unlabeled Data → Find Patterns → Create Groups

Clustering

```
from sklearn.cluster import KMeans
```

```
import numpy as np
```

```
X = np.array([
```

```
    [1,2],
```

```
    [1,4],
```

Step

Example

```
[1,0],  
[10,2],  
[10,4],  
[10,0]  
])  
  
kmeans = KMeans(n_clusters=2)  
  
kmeans.fit(X)  
  
print(kmeans.labels_)
```

Output:

```
[1 1 1 0 0 0]
```

Difference Between Supervised and Unsupervised

Supervised	Unsupervised
Labeled data	Unlabeled data
Teacher available	No teacher
Predict output	Find hidden patterns
Regression, Classification	Clustering

Part 5: Hands-on Practical Activity 1: Student Marks Prediction

Task:

- Predict marks from study hours

```
import pandas as pd  
  
from sklearn.linear_model import LinearRegression  
  
data = {  
    "Hours": [1, 2, 3, 4, 5],  
    "Marks": [20, 40, 50, 70, 90]}
```

```
}  
  
df = pd.DataFrame(data)  
  
X = df[["Hours"]]  
  
y = df["Marks"]  
  
model = LinearRegression()  
  
model.fit(X,y)  
  
prediction = model.predict([[6]])  
  
print("Predicted Marks:", prediction)
```

Activity 2: Customer Grouping

```
from sklearn.cluster import KMeans
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
X = np.array([
```

```
    [1,2],
```

```
    [2,3],
```

```
    [3,4],
```

```
    [10,10],
```

```
    [11,11],
```

```
    [12,12]
```

```
])
```

```
kmeans = KMeans(n_clusters=2)
```

```
kmeans.fit(X)
```

```
plt.scatter(X[:,0], X[:,1], c=kmeans.labels_)
```

```
plt.show()
```

Explain:

- Similar customers grouped together
-

Part 6: Mini Project

Mini Project Ideas

Project	ML Type
House Price Prediction	Supervised
Student Result Prediction	Supervised
Customer Segmentation	Unsupervised
Spam Detection	Supervised

Example Project Flow

Student Result Prediction

Input:

- Study hours
- Attendance

Output:

- Marks prediction
-

Questions

1. What is Machine Learning?
2. Difference between AI and ML?
3. What is supervised learning?
4. What is unsupervised learning?
5. What is labeled data?
6. What is regression?
7. What is clustering?
8. Give real-life ML examples.
9. Explain ML workflow.
10. What is training data?

Task 1

Create:

- Student marks prediction model

Task 2

Find:

- 5 real-world ML applications

Task 3

Differentiate:

- Supervised vs Unsupervised learning

Final Summary

Students learned:

- Introduction to ML
- Real-world applications
- ML workflow
- Supervised learning
- Unsupervised learning
- Regression
- Clustering
- Practical implementation